Project: Worked examples
Title: Fillet weld (EN 1993-1-8)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the utilisation of a fillet weld loaded parallel, transverse and normal to its axis, with the required throat and leg. The simplified method compares the resultant over the throat with the design shear strength.

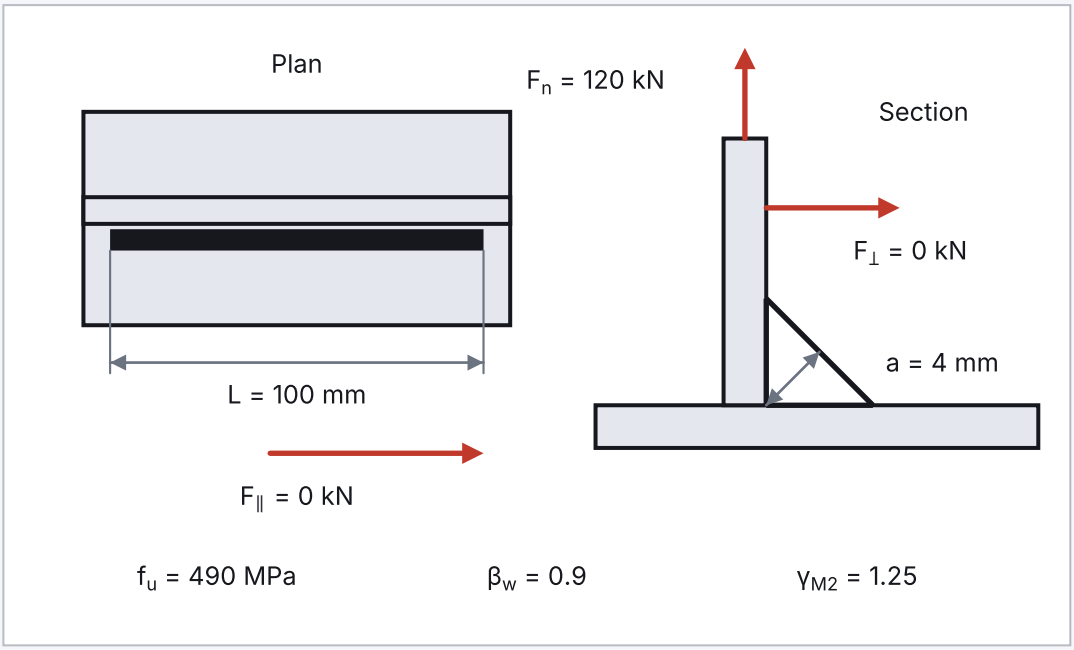


Figure 1 — A fillet weld of effective length L: in plan, the force along it; in section, the throat a and the forces across it and normal to the plate. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Resultant force on the weld	$F = \sqrt{(F_\parallel^2 + F_\perp^2 + F_n^2)}$
Design shear strength of the weld, simplified method	$f_{vw,d} = f_u/(\sqrt{3}\cdot\beta_w\cdot\gamma_{M2})$
Long-joint reduction of a lap joint longer than 150·a (4.11(3))	$\beta_{Lw} = \min(1, 1.2 - 0.2\cdot L/(150\cdot a))$; 1 for a shorter lap or another joint type
Shear stress on the throat	$\tau_w = F/(a\cdot L)$
Weld resistance and utilisation	$F_{w,Rd} = \beta_{Lw}\cdot f_{vw,d}\cdot a\cdot L$; $u = F/F_{w,Rd}$
Throat needed (not less than 3 mm, with the β_{Lw} of the throat provided) and the matching leg	$a_{req} = \max(F/(\beta_{Lw}\cdot f_{vw,d}\cdot L), 3\text{ mm})$; leg = $\sqrt{2}\cdot a$

Basis: Simplified method for design resistance of fillet weld[1](cl. 4.5.3.3); Effective throat thickness: minimum 3 mm[1](cl. 4.5.2(2)); Effective length: welds shorter than 30 mm or 6 throats not to carry load[1](cl. 4.5.1(2)); Long joints[1](cl. 4.11).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
joint	Joint type: a lap joint is reduced beyond 150 throats, and Other is for a weld whose stress follows the base metal	Lap joint		assumption: Lap joint (template default); the weld is shorter than 150 throats, so the long-joint reduction does not apply.	any
L	Effective weld length	100	mm	drawing: DRG-WD-702 rev A	30 … 10000 mm
a	Throat thickness	4	mm	drawing: DRG-WD-702 rev A	3 … 50 mm
f_u	Ultimate tensile strength of the weaker part joined	490	MPa	spec: SPEC-STR-02 design criteria	300 … 1000 MPa
F_{par}	Force parallel to the weld axis	0	kN	spec: LOAD-SCHEDULE-01	-100000 … 100000 kN
F_{tr}	Force transverse to the weld axis, in the plane of the parts	0	kN	spec: LOAD-SCHEDULE-01	-100000 … 100000 kN
F_n	Force normal to the plane of the parts	120	kN	spec: LOAD-SCHEDULE-01	-100000 … 100000 kN
β_w	Correlation factor (0.8 S235, 0.85 S275, 0.9 S355, 1.0 S420 and S460)	0.9		spec: SPEC-STR-02 design criteria	0.8 … 1
γ_{M2}	Partial factor for the weld resistance	1.25		spec: SPEC-STR-02 design criteria	1 … 2

3. Assumptions

A1. The resultant force is shared uniformly along the effective length. (Simplified method.)
A2. Equal-leg fillet: leg = √2 × throat. (Geometry.)
A3. A lap joint longer than 150·a is reduced by $\beta_{Lw} = 1.2 - 0.2 \cdot L/(150 \cdot a)$, not above 1. (BS EN 1993-1-8 4.11(3).)
A4. The weld length L is the length of the lap in the direction of the force. (BS EN 1993-1-8 4.11(3).)
A5. joint (Joint type: a lap joint is reduced beyond 150 throats, and Other is for a weld whose stress follows the base metal) is taken as Lap joint. (Lap joint (template default); the weld is shorter than 150 throats, so the long-joint reduction does not apply.)

4. Working

[10] β_{Lw} = no long-joint reduction [1](4.11)
= lap not longer than 150·a
= 1
Long-joint reduction factor (1: not reduced)

[11] $F = \mathrm{sqrt}(F_{par}\text{^}2 + F_{tr}\text{^}2 + F_n\text{^}2)$
= sqrt((0 kN)^2 + (0 kN)^2 + (120 kN)^2)
= 120 kN
Resultant force on the weld

[12] $f_{vw,d} = f_u / (\sqrt{3} * \beta_w * \gamma_{M2})$ [1](4.5.3.3)
$= 490\ \text{MPa} / (\sqrt{3} \times 0.9 \times 1.25)$
$= 251.47\ \text{MPa}$
Design shear strength of the weld
[13] $\tau_w = F / (a * L)$
$= 120\ \text{kN} / (4\ \text{mm} \times 100\ \text{mm})$
$= 300\ \text{MPa}$
Stress on the throat
[14] $F_{w,Rd} = \beta_{Lw} * f_{vw,d} * a * L$ [1](4.5.3.3)
$= 1 \times 251.47\ \text{MPa} \times 4\ \text{mm} \times 100\ \text{mm}$
$= 100.59\ \text{kN}$
Design resistance of the weld, reduced by the long-joint factor when it applies
[15] $u = F / F_{w,Rd}$
$= 120\ \text{kN} / 100.59\ \text{kN}$
$= 1.193$
Utilisation
[16] $a_{calc} = F / (\beta_{Lw} * f_{vw,d} * L)$
$= 120\ \text{kN} / (1 \times 251.47\ \text{MPa} \times 100\ \text{mm})$
$= 4.772\ \text{mm}$
Throat needed for the load (with the long-joint factor of the throat provided)
[17] $a_{req} = \max(a_{calc}, 3\ \text{mm})$ [1](4.5.2(2))
$= \max(4.772\ \text{mm}, 3\ \text{mm})$
$= 4.772\ \text{mm}$
Required throat (not less than 3 mm)
[18] $s_{leg} = \sqrt{2} * a$
$= \sqrt{2} \times 4\ \text{mm}$
$= 5.6569\ \text{mm}$
Leg length of the throat provided (equal legs)

5. Results

SYMBOL	VALUE	DESCRIPTION
$F_{w,Rd}$	100.58 kN	Weld resistance
u	1.20	Utilisation
$f_{vw,d}$	251.5 MPa	Design shear strength
τ_w	300 MPa	Throat stress
a_{req}	4.772 mm	Required throat
s_{leg}	5.657 mm	Leg length provided
F	120 kN	Resultant force (governing)
β_{Lw}	1	Long-joint factor
a_{calc}	4.772 mm	Throat needed

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Weld force against weld resistance	120 kN	100.58 kN	1.20	[1](4.5.3.3)

7. Applicability

None.

8. Verification

Template struct.fillet-weld@1.2.0 Content hash 26d0417a5e3ec5c6 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-8: Design of Joints. BS EN 1993-1-8:2005. BSI; 2005.